

Perimeter, Area & Volume

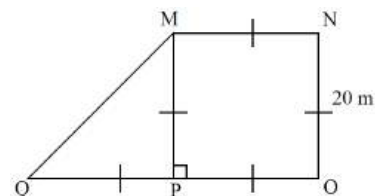
Topic Test 1 Solution

Total Marks: 30

Time: 50 minutes

- 1 What is the perimeter of the shape $MNOQ$, correct to the nearest metre?

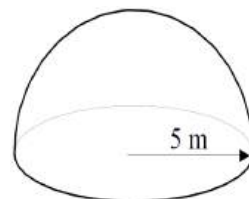
$$\sqrt{20^2 + 20^2} + (20 \times 4) = 108$$



1

- 2 Calculate the volume of a hemisphere with a radius of 5 m. Answer correct to one decimal place.

$$\frac{4}{3} \times \pi \times 5^3 \times \frac{1}{2} = 261.8 \text{ m}^3$$



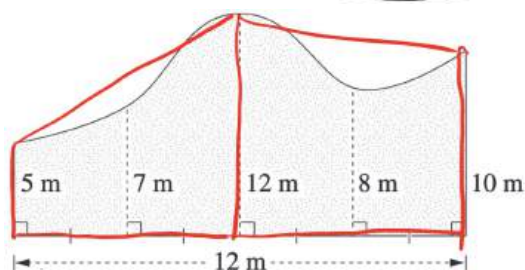
1

- 3 The diagram represents a field:

Find the area of the field, using two applications of the trapezoidal rule.

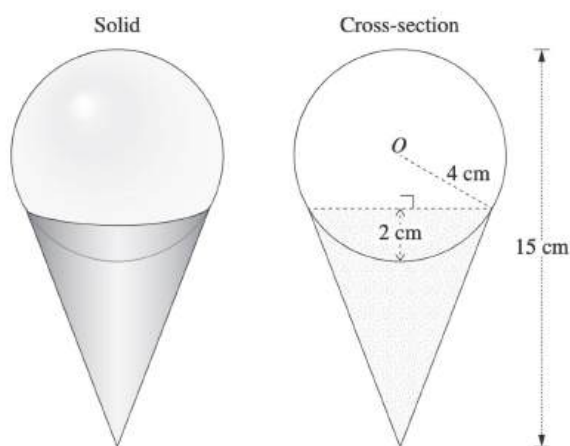
$$n = 6$$

$$A \approx \frac{6}{2} (5 + 12) + \frac{6}{2} (12 + 10) = 117 \text{ m}^2$$



2

- 4 A solid is made up of a sphere sitting partially inside a cone. The sphere, centre O, has a radius of 4 cm and sits 2 cm inside the cone. The solid has a total height of 15 cm. The solid and its cross-section are:



3

What is the volume of the cone, correct the nearest cubic centimetre?

$$\text{radius of cone} \rightarrow r^2 = 4^2 - 2^2 \quad r = \sqrt{12}$$

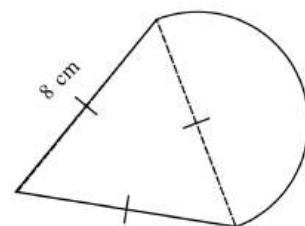
$$\text{Volume of cone} \rightarrow \frac{1}{3} \pi r^2 h = \frac{1}{3} \pi \times (\sqrt{12})^2 \times 9 = 113 \text{ cm}^3$$

- 5 Calculate the perimeter of this shape consisting of a semicircle and an equilateral triangle, correct to 1 decimal place.

$$l = \frac{\theta}{360} \times 2\pi r$$

$$= \frac{180}{360} \times 2 \times \pi \times 4 = 12.56637061$$

$$P = 2 \times 8 + 12.566 = 28.6 \text{ cm (to 1 d. p.)}$$



2

- 6 The diagram shows a can of Yummy soup and its label. Calculate the radius of the can, correct to 1 decimal place.



2

$$C = 2 \times \pi \times r$$

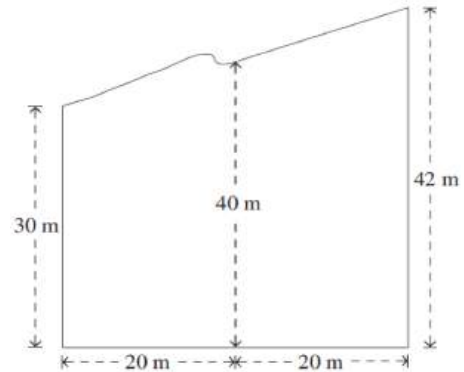
$$2 \times \pi \times r = 23.6$$

$$r = 3.7560\dots = 3.8 \text{ cm}$$

- 7 The diagram below shows the aerial view of a roof for a farm shed.

- a. Using two applications of the Trapezoidal rule, find the approximate area of the roof.

$$V \approx \left[\frac{20}{2} (30 + 40) + \frac{20}{2} (40 + 42) \right] = 1520 \text{ m}^2$$

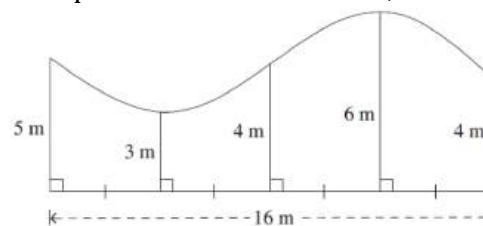


2

- b. Any rain that falls on to the roof flows directly into a rainwater tank. Assuming there is no runoff, calculate how many litres of water is collected in the rainwater tank when 120 mm of rain falls onto the roof. $1 \text{ m}^3 = 1 \text{ kL}$

$$\text{Capacity} = 1520 \times 0.12 \times 1000 = 182\,400 \text{ L}$$

- 8 Luka is installing a pond in his backyard. The pond will have three straight edges and one curved edge. The depth of the pond will be 2 metres throughout. Luka took measurements and drew an overhead view of the pond to assist the builder, as shown in the diagram.



- a. Using four applications of the trapezoidal rule, estimate the area of the bottom of the pond.

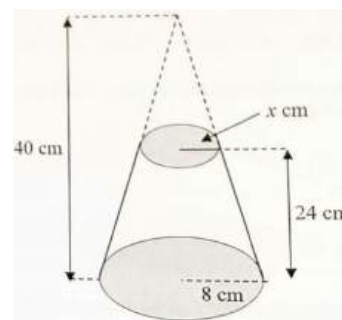
$$A \approx \frac{4}{2} \left[(5+3) + (3+4) + (4+6) + (6+4) \right] \approx 70 \text{ m}^2$$

- b. Calculate how many litres of water will be required to fill the pond.

$$70 \times 2 = 140 \text{ m}^3 \quad \times 1000 \quad 1 \text{ m}^3 = 1000 \text{ L} \\ = 140\,000 \text{ L}$$

2

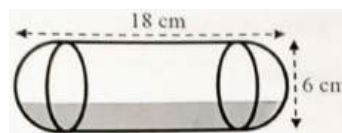
- 9 The top section of a solid cone of height 40 cm and base radius 8 cm is cut off leaving what is referred to as a “frustum” of height 24 cm, as shown in the diagram below. What is the length of the radius (x) of the top circular face of the frustum?



2

$$\frac{40 - 24}{8} = \frac{16}{40} \quad x = 3.2 \text{ cm} \quad (\text{similar triangles})$$

- 10 A capsule of length 18 cm and diameter 6 cm is formed by joining an open hemisphere at each end of an open cylinder, as shown in the diagram. Water is poured into the capsule and fills one third of its capacity.



- a. Show that the volume of water in the capsule is $48\pi \text{ cm}^3$.

3

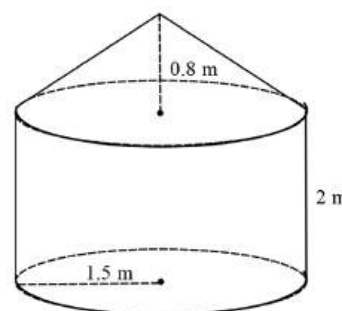
$$\begin{aligned} V &= \left(\frac{4}{3} \times 3^3 \times \pi \right) + (\pi \times 3^2 \times 12) \\ &= 36\pi + 108\pi \\ &= 144\pi \div 3 = 48\pi \text{ cm}^3 \end{aligned}$$

- b. The capsule is rotated 90° so that it sits vertically on one of its hemispherical ends, as shown in the following diagram. What is the height (h) correct to one decimal place, of the water level in the capsule?

3

$$\begin{aligned} 48\pi &= \left(\frac{1}{2} \times \frac{4}{3} \times 3^3 \times \pi \right) + (\pi \times 3^2 \times h) \\ &= 18\pi + 9\pi h \\ 30\pi &= 9\pi h \\ h &= \frac{30\pi}{9\pi} = 3\frac{1}{3} \text{ cm} + 3 \text{ (radius)} \\ &= 6\frac{1}{3} \text{ cm} \end{aligned}$$

- 11 The Morris family have a rainwater tank which consists of a cylinder with a cone shaped top. The tank, including dimensions, is shown below. Water enters through the top of the cone and can fill all the way to the peak.



2

- a. Calculate the volume of the tank.

$$V = \pi \times 1.5^2 \times 2 + \frac{1}{3} \pi \times 1.5^2 \times 0.8 = 16.02212253 \text{ m}^3$$

- b. Find the capacity of the tank, rounded to the nearest litre.

1

$$\begin{aligned} \text{Capacity} &= 16.02212253 \times 1000 \\ &= 16022.12253 \approx 16022 \text{ L} \end{aligned}$$

Perimeter, Area & Volume

Topic Test 2 Solution

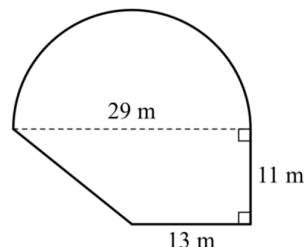
Total Marks: 30

Time: 50 minutes

- 1 The composite shape below is made up of a semicircle and a trapezium.

3

Calculate the perimeter of the shape, correct to the nearest metre.

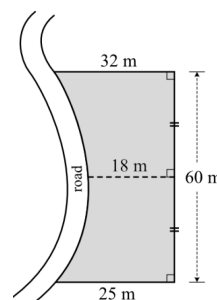


$$\begin{aligned}
 r &= 29 \div 2 = 14.5 \text{ m} \\
 \text{arc length} &= \frac{1}{2} \times 2 \times \pi \times 14.5 \\
 &= 14.5\pi \text{ m} \quad (45.55... \text{ m}) \\
 \text{Length of sloping side} & \quad \begin{array}{c} 16 \text{ m} \\ \text{ } \\ c \\ \text{ } \\ 11 \text{ m} \end{array} \\
 c^2 &= 16^2 + 11^2 \quad (\text{Pythagoras}) \\
 &= 377 \\
 c &= \sqrt{377} \text{ m} \quad (19.41... \text{ m}) \\
 \text{Perimeter} &= 14.5\pi + \sqrt{377} + 13 + 11 \\
 &= 88.969... \text{ m} = 89 \text{ m} \quad (\text{nearest metre})
 \end{aligned}$$

- 2 The shaded region represents a block of land bounded on one side by a road.

1

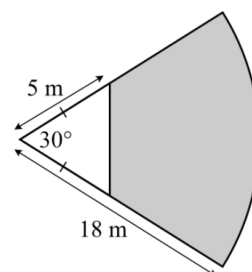
What is the approximate area of the block of land using two applications of the Trapezoidal rule?



$$\begin{aligned}
 h &= \frac{60}{2} = 30 \text{ m} \\
 A &\approx \frac{30}{2} (32 + 18) + \frac{30}{2} (18 + 25) \\
 &\approx 750 + 645 \approx 1395 \text{ m}^2
 \end{aligned}$$

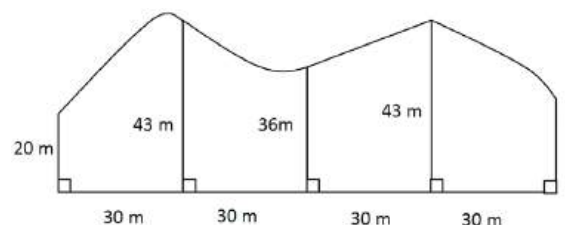
- 3 Calculate the area of the shaded region in the sector below, correct to two decimal places.

3



$$\begin{aligned}
 \text{Area of sector} &= \frac{30}{360} \times \pi \times 18^2 \\
 &= 27\pi \text{ m}^2 \quad (84.82... \text{ m}^2) \\
 \text{Area of triangle} &= \frac{1}{2} \times 5 \times 5 \times \sin 30^\circ \\
 &= 6.25 \text{ m}^2 \\
 \text{shaded area} &= 27\pi - 6.25 \\
 &= 78.573... \\
 &= 78.57 \text{ m}^2 \quad (2 \text{ dp})
 \end{aligned}$$

- 4 a. When using the Trapezoidal rule the area of the following shape was found to be 4335 m^2 . Find the value of x by using the trapezoidal rule four times.



2

$$4335 = 30/2(20+43) + 30/2(43 + 36) + 30/2(36 + 43) + 30/2(43 + x)$$

$$x = 25 \text{ m}$$

- b. The area above is to be fertilised. Ten litres of the fertiliser will cover 1000 m^2 . The fertiliser containers come in two sizes.

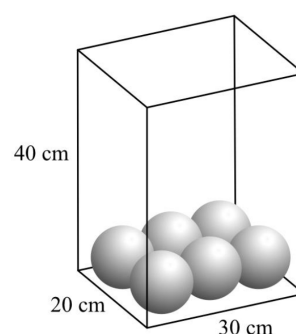
Size	Cost
10 litres	\$28.50
5 litres	\$16

2

Find the minimum cost needed to fertilise the area.

Containers that need to be purchased are four 10 litres and one 5 litre container.
cost = $4 \times \$28.50 + \$16 = \$130$

- 5 Solid spherical balls of diameter 10 cm are stacked inside a box as shown below. The box is in the shape of a rectangular prism such that 6 balls will fit snugly in each layer. The balls are stacked in the same manner as in the bottom layer until the box is full. Calculate the percentage of the box that is empty space. Give your answer correct to the nearest percentage.



3

Height of box is 40 cm and diameter of each ball is 10 cm $\therefore$ 4 layers of balls will fit.

number of balls = $4 \times 6 = 24$

Volume of balls = $24 \times \frac{4}{3} \times \pi \times 5^3$

$= 12566.370...$

$= 12566 \text{ cm}^3$ ✓ (nearest cm^3)

Volume of box = $20 \times 30 \times 40 = 24000 \text{ cm}^3$

Volume of space = $24000 - 12566 = 11434 \text{ cm}^3$

% space = $\frac{11434}{24000} \times 100 = 47.641 = 48\%$ ✓ (nearest %)

- 6 A cube has side length of 8.4 cm. What is the surface area of this cube?

423.36 cm^2

1

- 7 The formula for the volume of a cone is

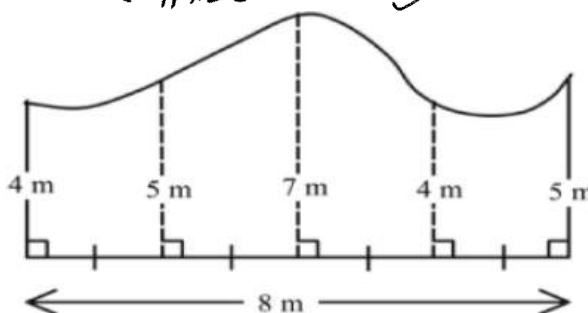
$$V = \frac{1}{3}\pi r^2 h$$

where V is the volume, h is the height and r is the radius.

- a. Show that the radius of the cone is $r = \sqrt{\frac{3V}{\pi h}}$. 2
- b. Calculate the radius of the base of a cone that is 20 cm high and has a volume of 200 cm³, correct to 2 decimal places. 2

$$\begin{aligned} 3V &= \pi r^2 h \\ \sqrt{\frac{3V}{\pi h}} &= r \\ \sqrt{\frac{3 \times 200}{\pi \times 20}} &= 3.09 \end{aligned}$$

- 8 A swimming pool has three straight sides and one which is curved. It is 1.5 metres deep at all points. Manjit takes measurements shown of the pool in preparation for filling it with water.



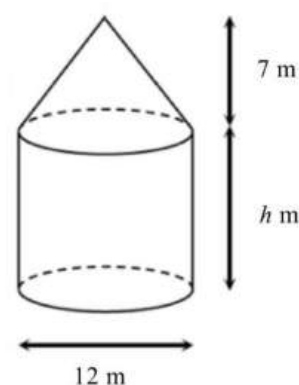
- a. Use the trapezoidal rule four times to estimate the area of the base of the pool. 2

$$A = \frac{2}{2} (4+5) + (5+7) + (7+4) + (4+5) = 41 \text{ m}^2$$

- b. Using the fact that 1 m³ holds 1000 litres of water, how many litres of water will be needed to fill the pool? 1

$$\begin{aligned} V &= 41 \times 1.5 = 61.5 \text{ m}^3 \\ \text{Capacity} &= 61.5 \times 1000 = 61500 \text{ L} \end{aligned}$$

- 9 The diagram below shows a silo made up of a cone with a perpendicular height of 7 m on top of a cylinder with a height of h m. The diameter of the cone and cylinder is 12 m. The volume of the silo is 1500 m³.



Determine the height of the cylinder h , correct to 3 significant figures.

$$\begin{aligned} \text{Radius} &= 6 \text{ m} \\ \text{Volume} &= \text{cone} + \text{cylinder} \\ \frac{1}{3}\pi(6)^2 7 + \pi(6)^2 h &= 1500 \\ 263.98 \dots + h \times 113.097 \dots &= 1500 \\ h \times 113.097 \dots &= 1236.10 \dots \\ h &= 10.929 \dots = 10.9 \text{ (3 sig fig)} \end{aligned}$$

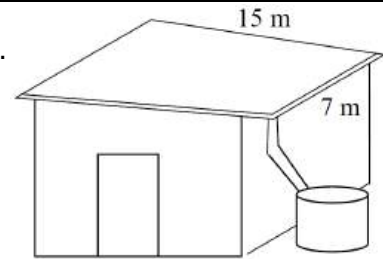
- 10 Earth is a sphere with a diameter of approximately 12 683 km. Four-fifths of the Earth's surface is covered by water. Calculate the area of the Earth's surface that is covered by water. Give your answer correct to the nearest thousand square kilometres. 2

$$A = 4\pi \left(\frac{12683}{2} \right)^2 \times \frac{4}{5} = 404\,281\,000 \text{ km}^2$$

- 11 A flat rectangular roof on a shed measures 15 metres by 7 metres.

Rain which falls on this roof, is collected in a cylindrical tank of height 3m and diameter 2.9 m. ($1000 L = 1 m^3$)

During a recent thunderstorm 89 millimetres of rain fell over a 24 hour period.



- a. What rise in water level in the tank would be caused by this rain? 3

$$\text{Volume of water on the roof} = 15 \times 7 \times 89/1000 = 9.345 \text{ m}^3$$

$$\begin{aligned} \text{Height of tank: } &= 9.345 = \pi \times \left(\frac{2.9}{2}\right)^2 \times h \\ h &= 4 \times 9.345 / (\pi \times 2.9^2) = 1.415 \text{ m} \end{aligned}$$

- b. If the tank is empty, how many millimetres of rain are needed to fall on the roof, to fill the tank? 2

$$15 \times 7 \times \frac{x}{1000} = \pi \left(\frac{2.9}{2}\right)^2 \times 3 \quad \text{where } x \text{ is the number of } mm \text{ needed to fill the tank}$$

$$x = 188.72 \text{ mm}$$